

5 Number theory	5.1	An understanding of the division theorem and its application to the Euclidean Algorithm and congruences.	Students should be able to apply the algorithm to find the highest common factor of two numbers.
	5.2	Bezout's identity.	Students should be able to use back substitution to identity the Bezout's identity for two numbers.
	5.3	Modular arithmetic. Understanding what is meant by two integers a and b to be congruent modulo an integer n. Properties of congruences.	The notation $a \equiv b \pmod{n}$ is expected. Knowledge of the following properties: $a \equiv a \pmod{n}$ if $a \equiv b \pmod{n}$ then $b \equiv a \pmod{n}$ if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$ Addition and subtraction laws for congruences. Multiplication and power laws.
	5.4	Fermat's Little Theorem.	For example, students should be able to find the least positive residue of 4^{20} modulo 7 Proof is not required.

Topics	What students need to learn:		
	Content		Guidance
5 Number theory <i>continued</i>	5.5	Divisibility Tests.	For divisibility by 2, 3, 4, 5, 6, 9, 10 and 11.
	5.6	Solution of congruence equations.	Conditions under which solutions exist should be known. Use of Bezout's identity to find multiplicative inverses.
	5.7	Combinatorics: counting problems, permutations and combinations.	The multiplicative principle of counting. Set notation is expected; the number of subsets of a set should be known. Addition and subtraction principles. Students should, for example, be able to determine the number of positive integers less than 1000 containing the digit 3 (at least once). Understanding both nP_r and nC_r and when to use them. For example, to determine how many ways there are to select a team of 11 players from a squad of 21 (a) regardless of playing positions and (b) if positions matter